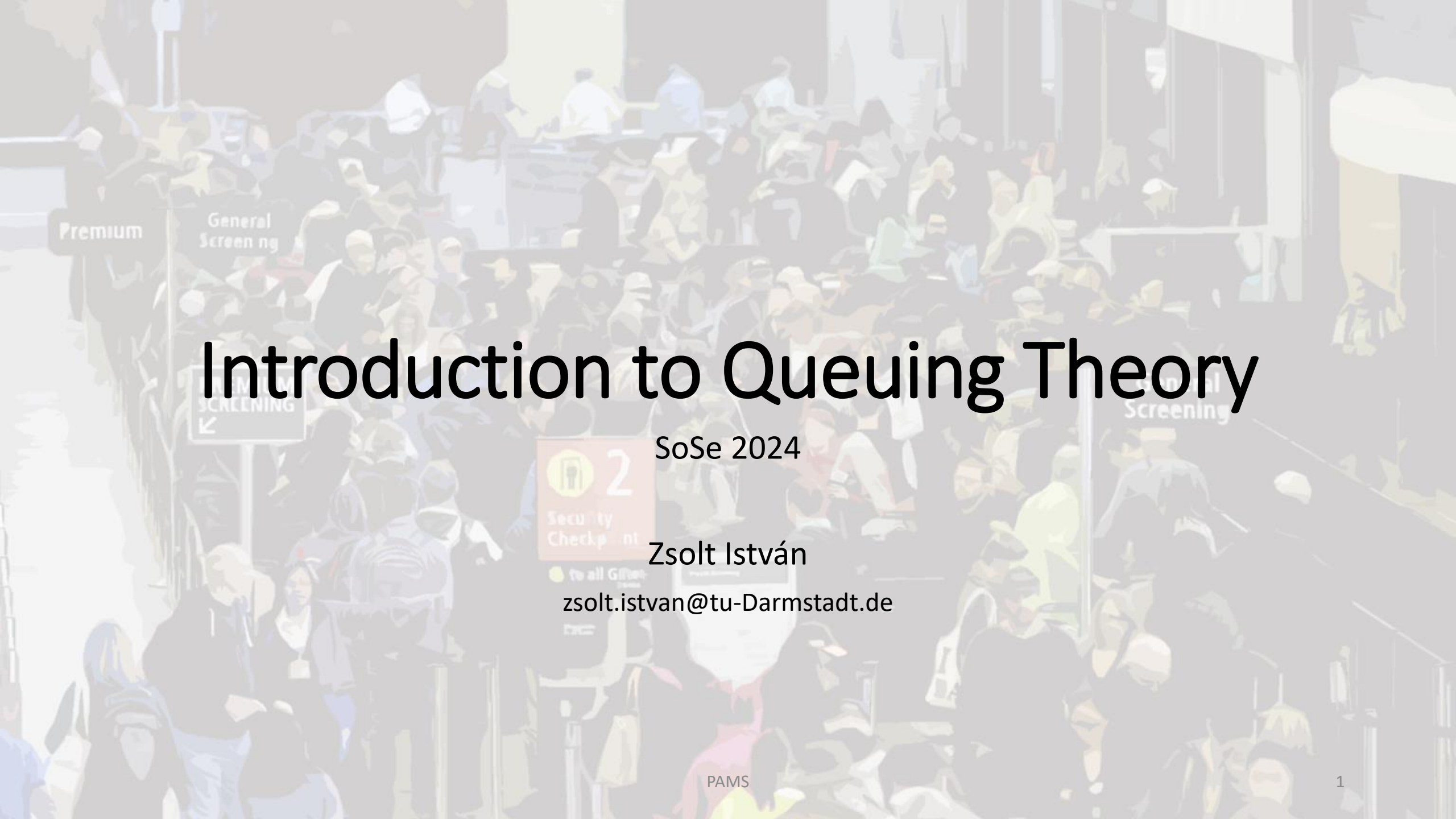




Introduction to Queuing Theory

SoSe 2024

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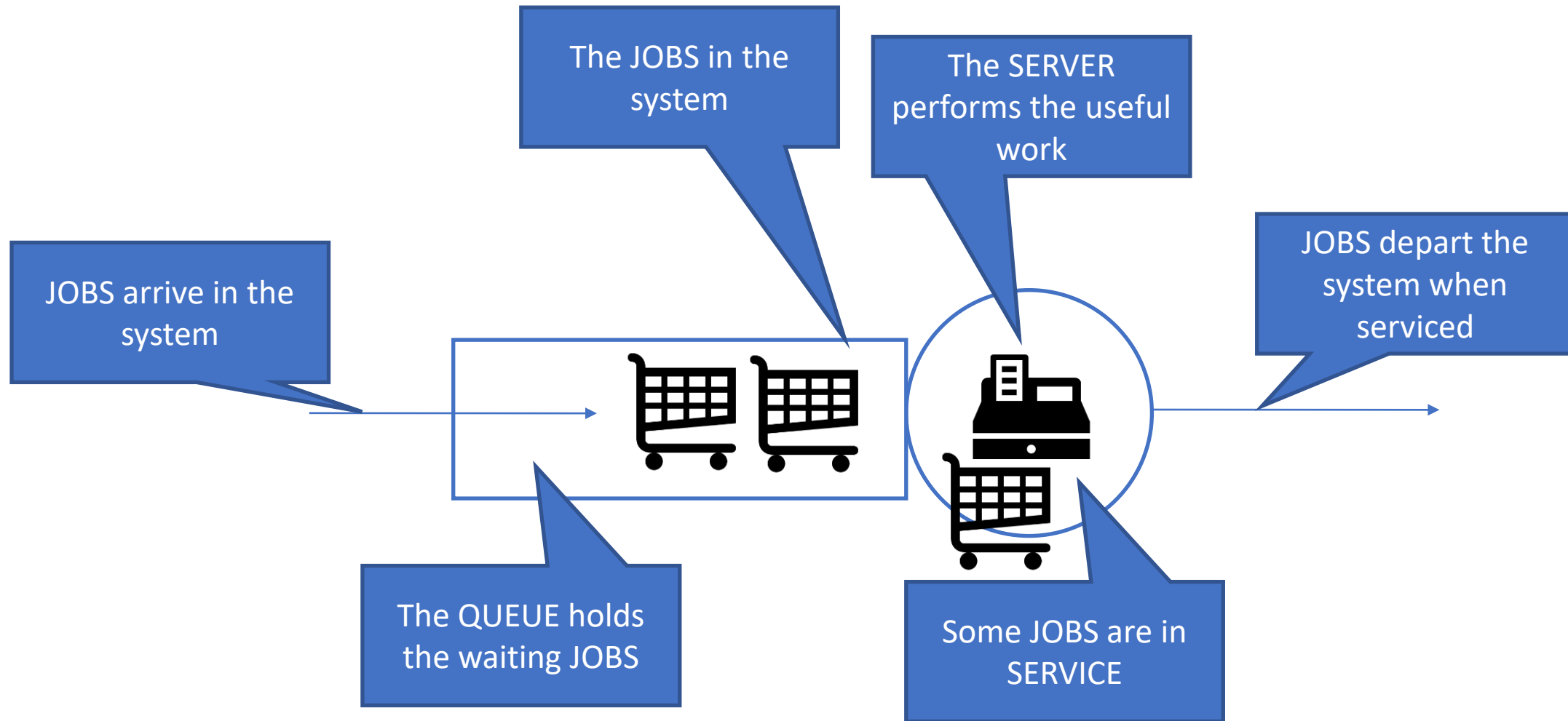
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Last weeks

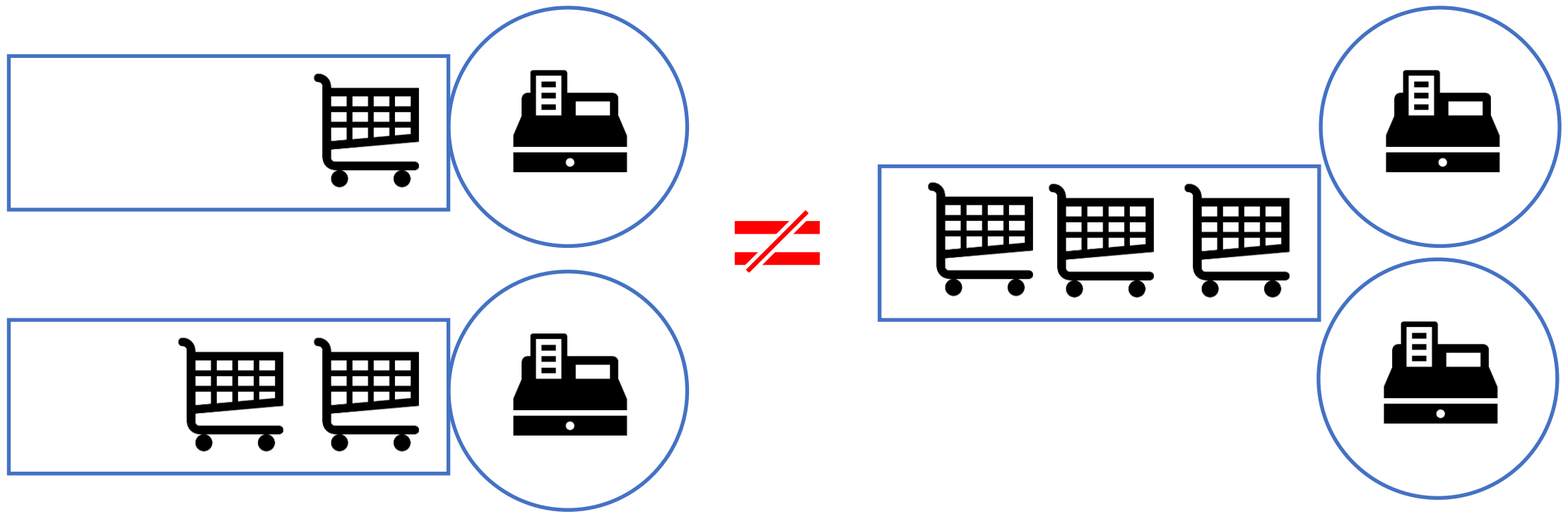
- Experimental Design Basics
- Accuracy/Precision
- Mean/Median and Percentiles
- Warm-up/Cool-down
- Open/Closed systems
- Operational laws and Interactive Response Time Law

What is a queuing system?

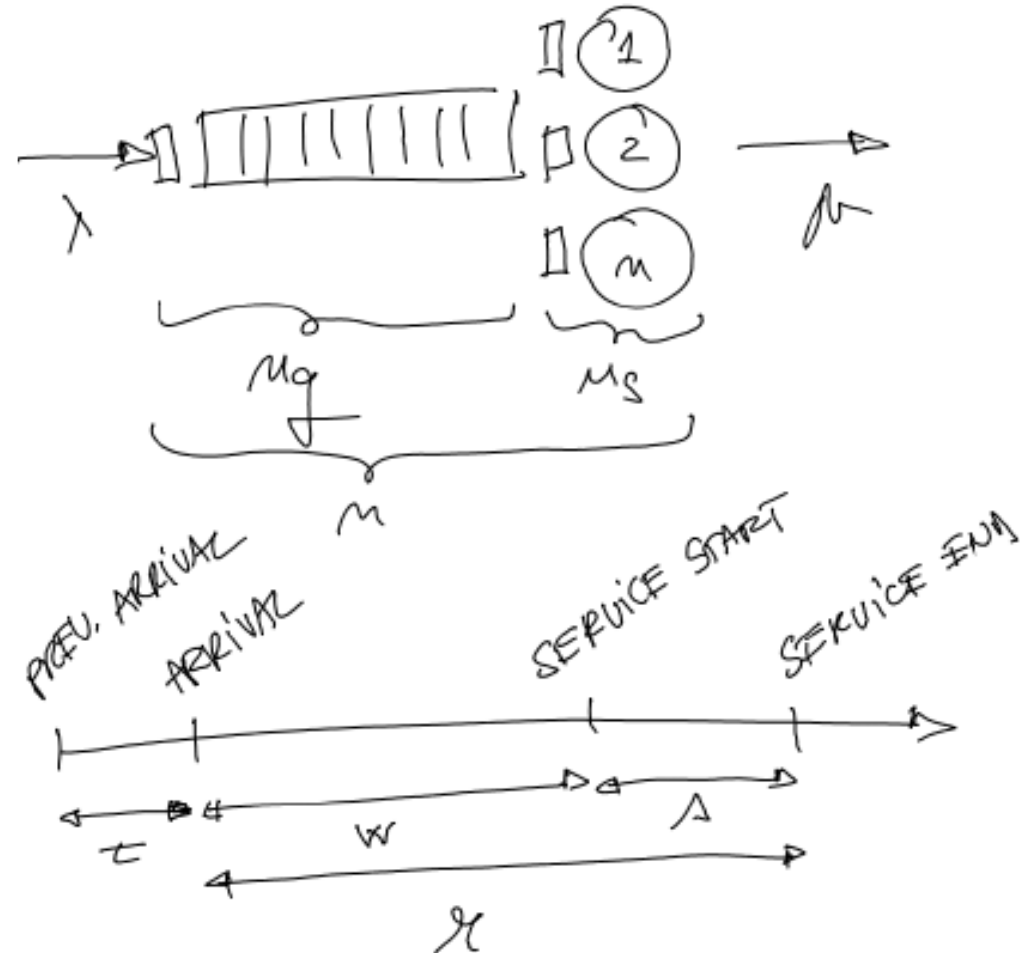
A 10.000ft view of a queueing system



Spoiler alert!



Queueing System in Numbers



Interarrival time

- Interarrival time τ (*tau*) is a random variable
 - Mean value: $E[\tau]$
- Typically assumed to be Poisson:
 - (1) the event is something that can be counted in whole numbers
 - (2) occurrences are independent, so that one occurrence neither diminishes nor increases the chance of another
 - (3) the average frequency of occurrence for the time period in question is known

Mean Arrival rate

- Mean arrival rate λ (*lambda*)

$$\lambda = 1/E[\tau]$$

- Useful to assume fixed λ for modeling
 - Do real systems have fixed λ ?
- Examples:
 - A new pizza order is received on average every 3 minutes. The arrival rate is 20/hour.
 - The printer receives a new job to print on average every 100ms, the arrival rate is 10jobs/second.

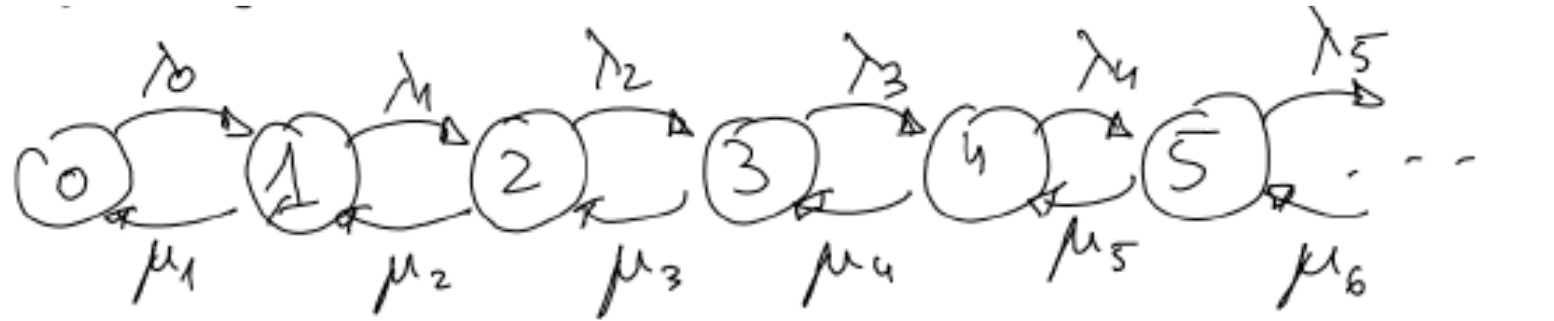
Service time

- Time to process a job (“useful work”, no queueing)
 - Random variable: s
- Mean service rate $\mu = 1 / E[s]$
 - What if we have m servers?
 - Not a random variable
- Example: pizza oven bakes pizza on average in 6 minutes. $\mu = 10/\text{hour}$

Number of jobs

- Number of jobs in the system is the sum of the jobs in the queue and the ones in service
 - $n = n_s + n_q$
- Total time spent in system (response time) is the sum of time spent queuing and that in service
 - $r = w_q + s$
- Remember these are random variables, we'll speak of their expected value.

Birth-Death Processes



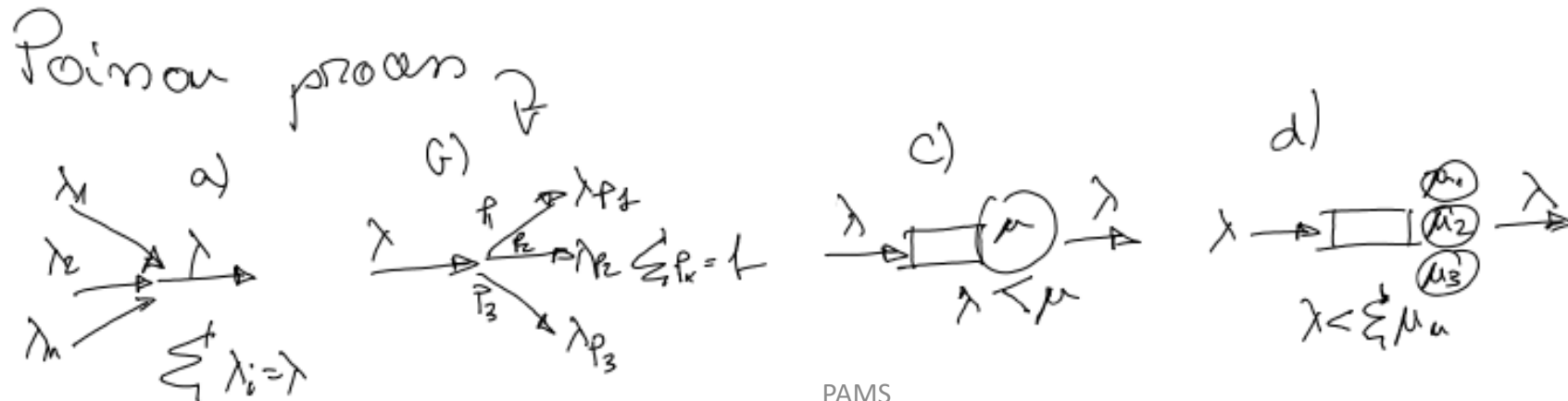
$$p_n = \frac{\lambda_0 \cdot \lambda_1 \cdot \lambda_2 \cdots \lambda_{n-1}}{\mu_1 \cdot \mu_2 \cdots \mu_n} p_0 \quad n = 1, 2, \dots, \infty$$

$\nwarrow$ PROBABILITY OF BEING IN STATE " n "

$\nwarrow$ PROBABILITY OF BEING IN ZERO STATE

Poisson Process

- Birth-death process with
 - Interarrival times independent and identically distributed variables
 - Interarrival times are exponentially distributed
 - Number of arrivals over period of time Poisson distributed
- Poisson processes have nice properties:



A word on throughput

- The service rate of a system is rarely the measured throughput!
 - Throughput is client and workload dependent
 - Throughput only counts successful operations
- Is arrival rate the same as throughput?
 - In open systems?
 - In closed systems?
 - If there are no failed jobs?

Default assumptions for other properties

- What is the service discipline?
 - First come, first served (FIFO)
- What is the system capacity?
 - Large enough buffers → Infinite buffers
- What is the population size?
 - Very large → Infinite size

See book for methodology on other types of queueing systems

Little's Law

- Holds for any system where departures = arrivals

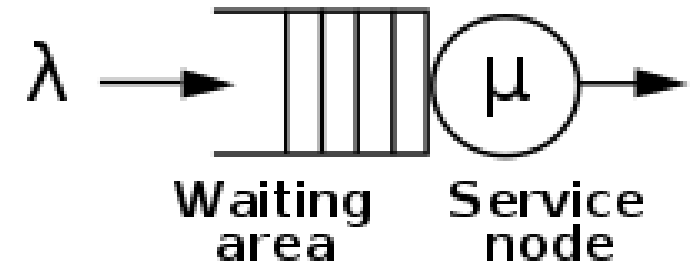
Mean number of jobs in system = arrival rate * mean response time

$$E[n] = \lambda * E[r]$$

- Recall: $n = n_s + n_q$
 - Only the queue part: $E[n_q] = \lambda * E[w_q]$
 - Only the service part: $E[n_s] = \lambda * E[s]$

Quick overview of M/M/1 queues

- Interarrival times and service times exponentially distributed (come from Poisson processes)
- Single server
- FIFO processing
- Parameters:
 - Mean arrival rate
 - Mean service rate



INTERARRIVAL TIME DISTRIBUTION: EXPONENTIAL
 $M/M/1$ Queue NUMBER OF SERVERS: 1
SERVICE TIME DISTRIBUTION: EXPONENTIAL

M/M/1 as Poisson process

M/M/1 Queue as Poisson process:

$$\lambda_n = \lambda \quad n = 0, 1, 2, \dots, \infty$$
$$\mu_n = \mu$$

$$P_n = \frac{\lambda_0 \cdot \lambda_1 \cdot \lambda_2 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \cdot P_0 = \left(\frac{\lambda}{\mu}\right)^n \cdot P_0 \quad n = 1, 2, \dots, \infty$$

$\rho = \frac{\lambda}{\mu}$ is TRAFFIC INTENSITY, $\mu > \lambda$ to BE POISSON PROCESS

Hint: sum of all probabilities = 1

$$P_0 = 1 - \rho$$

$$P_n = \rho^n \cdot (1 - \rho)$$

Other formulas

$$E[u] = \frac{\rho}{1-\rho}$$

$$E[n] = \frac{1}{\mu} \cdot \frac{1}{1-\rho}$$

Box 31.1 M/M/1 Queue

1. Parameters:

λ = arrival rate in jobs per unit time

μ = service rate in jobs per unit time

2. Traffic intensity: $\rho = \lambda/\mu$

3. Stability condition: Traffic intensity ρ must be less than 1.

4. Probability of zero jobs in the system: $p_0 = 1 - \rho$

5. Probability of n jobs in the system: $p_n = (1 - \rho)\rho^n$, $n = 0, 1, \dots, \infty$

6. Mean number of jobs in the system: $E[n] = \rho/(1 - \rho)$

7. Variance of number of jobs in the system: $\text{Var}[n] = \rho/(1 - \rho)^2$

8. Probability of k jobs in the queue:

$$P(n_q = k) = \begin{cases} 1 - \rho^2, & k = 0 \\ (1 - \rho)\rho^{k+1}, & k > 0 \end{cases}$$

9. Mean number of jobs in the queue: $E[n_q] = \rho^2/(1 - \rho)$

10. Variance of number of jobs in the queue:

$$\text{Var}[n_q] = \rho^2(1 + \rho - \rho^2)/(1 - \rho)^2$$

11. Cumulative distribution function of the response time:

$$F(r) = 1 - e^{-r\mu(1-\rho)}$$

12. Mean response time: $E[r] = (1/\mu)/(1 - \rho)$

13. Variance of the response time: $\text{Var}[r] = \frac{1/\mu^2}{(1 - \rho)^2}$

14. q -Percentile of the response time: $E[r] \ln[100/(100 - q)]$

15. 90-Percentile of the response time: $2.3E[r]$

16. Cumulative distribution function of waiting time:

$$F(w) = 1 - \rho e^{-\mu w(1-\rho)}$$

17. Mean waiting time: $E[w] = \rho \frac{1/\mu}{1 - \rho}$

18. Variance of the waiting time: $\text{Var}[w] = (2 - \rho)\rho/[\mu^2(1 - \rho)^2]$

19. q -Percentile of the waiting time: $\max\left(0, \frac{E[w]}{\rho} \ln[100\rho/(100 - q)]\right)$

20. 90-Percentile of the waiting time: $\max\left(0, \frac{E[w]}{\rho} \ln[10\rho]\right)$

21. Probability of finding n or more jobs in the system: ρ^n

22. Probability of serving n jobs in one busy period:

$$\frac{1}{n} \binom{2n-2}{n-1} \frac{\rho^{n-1}}{(1 + \rho)^{2n-1}}$$

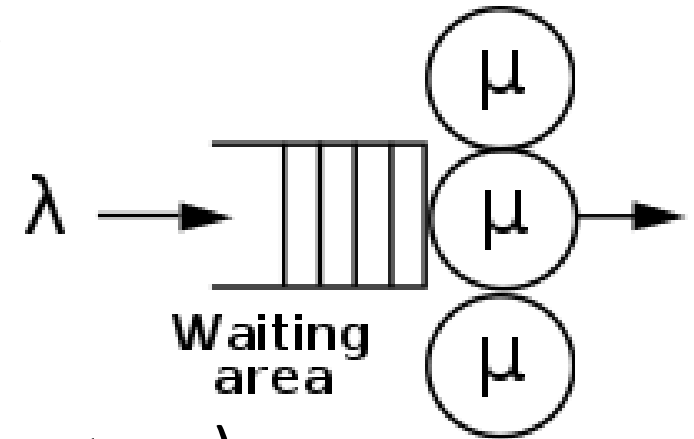
Some examples

- A USB thumb drive is serving 5k I/O ops/s. The average time seen by clients per I/O operation is 0.1ms. What is its utilization?
- When clients access a website, the page is rendered in 200ms. If we know that the webserver spends 10ms computing, how many requests are, on average, queued up in the system?

Are all servers M/M/1?

Quick overview of M/M/m queues

- Interarrival times and service times Poisson
 - FIFO processing
 - m parallel servers (no queueing if the number of jobs $\leq m$)
-
- Parameters:
 - Mean arrival rate
 - Mean service rate
 - Server parallelism



All formulas for M/M/1 and M/M/m

Box 31.1 M/M/1 Queue

- Parameters:
 λ = arrival rate in jobs per unit time
 μ = service rate in jobs per unit time
- Traffic intensity: $\rho = \lambda/\mu$
- Stability condition: Traffic intensity ρ must be less than 1.
- Probability of zero jobs in the system: $p_0 = 1 - \rho$
- Probability of n jobs in the system: $p_n = (1 - \rho)\rho^n$, $n = 0, 1, \dots, \infty$
- Mean number of jobs in the system: $E[n] = \rho/(1 - \rho)$
- Variance of number of jobs in the system: $\text{Var}[n] = \rho/(1 - \rho)^2$
- Probability of k jobs in the queue:

$$P(n_q = k) = \begin{cases} 1 - \rho^2, & k = 0 \\ (1 - \rho)\rho^{k+1}, & k > 0 \end{cases}$$

- Mean number of jobs in the queue: $E[n_q] = \rho^2/(1 - \rho)$
- Variance of number of jobs in the queue:
 $\text{Var}[n_q] = \rho^2(1 + \rho - \rho^2)/(1 - \rho)^2$
- Cumulative distribution function of the response time:
 $F(r) = 1 - e^{-r\mu(1-\rho)}$
- Mean response time: $E[r] = (1/\mu)/(1 - \rho)$
- Variance of the response time: $\text{Var}[r] = \frac{1/\mu^2}{(1 - \rho)^2}$
- q -Percentile of the response time: $E[r] \ln[100/(100 - q)]$
- 90-Percentile of the response time: $2.3E[r]$
- Cumulative distribution function of waiting time:
 $F(w) = 1 - \rho e^{-\mu w(1-\rho)}$
- Mean waiting time: $E[w] = \rho \frac{1/\mu}{1 - \rho}$
- Variance of the waiting time: $\text{Var}[w] = (2 - \rho)\rho/[\mu^2(1 - \rho)^2]$
- q -Percentile of the waiting time: $\max\left(0, \frac{E[w]}{\rho} \ln[100\rho/(100 - q)]\right)$
- 90-Percentile of the waiting time: $\max\left(0, \frac{E[w]}{\rho} \ln[10\rho]\right)$
- Probability of finding n or more jobs in the system: ρ^n
- Probability of serving n jobs in one busy period:

$$\frac{1}{n} \binom{2n-2}{n-1} \frac{\rho^{n-1}}{(1 + \rho)^{2n-1}}$$

Box 31.2 M/M/m Queue

- Parameters:
 λ = arrival rate in jobs per unit time
 μ = service rate in jobs per unit time
 m = number of servers
- Traffic intensity: $\rho = \lambda/(m\mu)$
- The system is stable if the traffic intensity ρ is less than 1.
- Probability of zero jobs in the system:

$$p_0 = \left[1 + \frac{(m\rho)^m}{m!(1-\rho)} + \sum_{n=1}^{m-1} \frac{(m\rho)^n}{n!} \right]^{-1}$$

- Probability of n jobs in the system:

$$p_n = \begin{cases} \frac{p_0 (m\rho)^n}{n!}, & n < m \\ \frac{p_0 \rho^n m^m}{m!}, & n \geq m \end{cases}$$

- Probability of queueing:

$$\rho = P(\geq m \text{ jobs}) = \frac{(m\rho)^m}{m!(1-\rho)} p_0$$

In the remaining formulas below we will use ρ as defined here.

- Mean number of jobs in the system: $E[n] = m\rho + \rho\rho/(1 - \rho)$
- Variance of number of jobs in the system:

$$\text{Var}[n] = m\rho + \rho\rho \left[\frac{1 + \rho - \rho\rho}{(1 - \rho)^2} + m \right]$$

- Mean number of jobs in the queue: $E[n_q] = \rho\rho/(1 - \rho)$
- Variance of number of jobs in the queue:
 $\text{Var}[n_q] = \rho\rho(1 + \rho - \rho\rho)/(1 - \rho)^2$
- Average utilization of each server: $U = \lambda/(m\mu) = \rho$
- Cumulative distribution function of response time:

$$F(r) = \begin{cases} 1 - e^{-\mu r} - \frac{\rho}{1 - m + m\rho} e^{-m\mu(1-\rho)r} - e^{-\mu r}, & \rho \neq (m-1)/m \\ 1 - e^{-\mu r} - \rho\mu r e^{-\mu r}, & \rho = (m-1)/m \end{cases} \quad r > 0$$

Box 31.2 Continued

- Mean response time:

$$E[r] = \frac{1}{\mu} \left(1 + \frac{\rho}{m(1-\rho)} \right)$$

- Variance of the response time:

$$\text{Var}[r] = \frac{1}{\mu^2} \left[1 + \frac{\rho(2-\rho)}{m^2(1-\rho)^2} \right]$$

- Cumulative distribution function of waiting time:

$$F(w) = 1 - \rho e^{-m\mu(1-\rho)w}$$

- Mean waiting time: $E[w] = E[n_q]/\lambda = \rho/[m\mu(1-\rho)]$
- Variance of the waiting time: $\text{Var}[w] = \rho(2-\rho)/[m^2\mu^2(1-\rho)^2]$
- q -Percentile of the waiting time: $\max\left(0, \frac{E[w]}{\rho} \ln \frac{100\rho}{100-q}\right)$
- 90-Percentile of the waiting time: $\frac{E[w]}{\rho} \ln(10\rho)$

Once again, ρ in these formulas is the probability of m or more jobs in the system: $\rho = [(m\rho)^m / \{m!(1-\rho)\}] p_0$. For $m = 1$, ρ is equal to ρ and all of the formulas become identical to those for M/M/1 queues.

Exercise

- $\mu = 250/s$
- $\lambda = 1200/s$
- What is better for clients ($E[r]$): **5x M/M/1** or **1x M/M/5**?
- For Wolfram Alpha:
- “Prob of 0 jobs in system” =
 - solve $p_0 = (1 / (1 + ((m * \rho)^m) / (m! * (1 - \rho)) + (\sum ((m * \rho)^n) / n!, n = 1..m - 1)))$, $m = 5$, $\rho = 1200 / (5 * 250)$
- “Prob of queueing” =
 - solve $q = (m \rho)^m / (m! (1 - \rho)) * p_0$; $m = 5$; $p_0 = 0.00169$; $\rho = 0.96$
- “Expected response time (needs prob of queueing)”
 - solve $x = (1 / \mu * (1 + q / (m * (1 - \rho))))$; $m = 5$; $\mu = 250$; $\rho = 0.96$; $q = 0.897$ for x

Exercise

- $\mu = 250/s$
- $\lambda = 1200/s$
- What is better for clients? **5x M/M/1** or **1x M/M/m**?
- For M/M/1: $E[r] = 0.1s$
- For M/M/m: $E[r] = 0.022s$
 - The jobs wait in a single queue and can go to any available server. In the other case they need to wait for their pre-chosen server to become available...

Hint: there are many tools/websites that help with computing the outputs of the models (e.g., <https://www.omnicalculator.com/math/queueing-theory>)

Homework – we'll discuss next week

1. For an M/M/1 queue with $\mu = 3500/\text{s}$, plot what is the expected response time as a function of utilization.
 - Assuming that each job is 1KB in size, how much memory should this server have for buffering jobs that are waiting for service?
2. For an M/M/7 queue with $\mu = 500/\text{s}$, plot what is the expected response time as a function of utilization.
 - Assuming that each job is 1KB in size, how much memory should this server have for buffering jobs that are waiting for service?
3. If we have an M/M/1 queue with $E[s]=5,7\text{ms}$. If we want to guarantee that, on average, requests finish in less than 50ms on this system, what is the higher utilization we can permit?
 - Hint: Make a plot of response time as a function of utilization and see where it crosses 50ms